

22.5 Interpreting a transformed model

A model fitted on a transformed scale answers questions on that scale, but readers ask about the original one: how much will a household with income 2000 spend on food, within what range, and how much more if its income rises? The dividing line is simple. Quantiles pass through a monotone transformation unchanged; means do not.

Throughout this section g is strictly increasing with inverse h , and the model on the transformed scale is

$$g(Y_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \dots, n, \quad (22.5.1)$$

where $\varepsilon_1, \dots, \varepsilon_n$ are independent with distribution F , mean 0 and variance σ^2 .

22.5.1 Medians and means

Proposition 22.5.1 — Retransformation bias and the smearing estimate

Assume Eq. 22.5.1.

- (a) **Quantiles.** If ξ_q is a q -quantile of F , then $h(\mathbf{x}^\top \boldsymbol{\beta} + \xi_q)$ is a q -quantile of Y at $\mathbf{x}$. In particular, if F has median 0 (for instance, if it is symmetric), the median of Y at $\mathbf{x}$ is $h(\mathbf{x}^\top \boldsymbol{\beta})$.
- (b) **Means.** $E(Y | \mathbf{x}) = \int h(\mathbf{x}^\top \boldsymbol{\beta} + e) dF(e)$ whenever this is finite. If h is strictly convex and $\sigma^2 > 0$, then $E(Y | \mathbf{x}) > h(\mathbf{x}^\top \boldsymbol{\beta})$. For $h = \exp$, $E(Y | \mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\beta}) E e^\varepsilon$, and for normal errors $E e^\varepsilon = \exp(\sigma^2/2)$.
- (c) **Smearing.** Let $h = \exp$, let $\mathbf{X}$ have full column rank p for every n , with leverages h_{ii} , and let the ε_i be the first n terms of an independent and identically distributed sequence with $E e^\varepsilon < \infty$. If $\max_i h_{ii} \rightarrow 0$ as $n \rightarrow \infty$, then the **smearing factor** satisfies

$$\frac{1}{n} \sum_{i=1}^n \exp(\hat{\varepsilon}_i) \rightarrow E e^\varepsilon \quad \text{in probability.} \quad (22.5.2)$$

If moreover $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \rightarrow \mathbf{x}_0^\top \boldsymbol{\beta}$ in probability, the **smearing estimate** $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) n^{-1} \sum_i \exp(\hat{\varepsilon}_i)$ converges in probability to $E(Y | \mathbf{x}_0)$.

Proof.

- (a) $Y \leq h(\mathbf{x}^\top \boldsymbol{\beta} + \xi_q)$ iff $g(Y) \leq \mathbf{x}^\top \boldsymbol{\beta} + \xi_q$ iff $\varepsilon \leq \xi_q$, because g is strictly increasing, and likewise with strict inequalities. So the two events have the probabilities that define a q -quantile.
- (b) $Y = h(\mathbf{x}^\top \boldsymbol{\beta} + \varepsilon)$. If h is strictly convex and ε is not degenerate, Jensen's inequality is strict: $E h(\mathbf{x}^\top \boldsymbol{\beta} + \varepsilon) > h(\mathbf{x}^\top \boldsymbol{\beta} + E \varepsilon) = h(\mathbf{x}^\top \boldsymbol{\beta})$. For $h = \exp$ the factor $\exp(\mathbf{x}^\top \boldsymbol{\beta})$ comes out of the expectation, and the normal moment generating function (Theorem 3.2.1) at 1 is $\exp(\sigma^2/2)$.

- (c) Put $\mathbf{d} = \mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{M}\boldsymbol{\varepsilon}$, so that $\hat{\varepsilon}_i = \varepsilon_i - d_i$. With $\mathbf{e}_i$ the i th standard basis vector, $d_i = \mathbf{e}_i^\top \mathbf{M}\boldsymbol{\varepsilon} = (\mathbf{M}\mathbf{e}_i)^\top (\mathbf{M}\boldsymbol{\varepsilon})$, so by the Cauchy–Schwarz inequality and Proposition 6.8.1(a),

$$|d_i| \leq \|\mathbf{M}\mathbf{e}_i\| \|\mathbf{M}\boldsymbol{\varepsilon}\| = \sqrt{h_{ii}} \|\mathbf{M}\boldsymbol{\varepsilon}\|.$$

Since $E\|\mathbf{M}\boldsymbol{\varepsilon}\|^2 = \sigma^2 \text{tr}(\mathbf{M}) = p\sigma^2$, Markov’s inequality gives $\Pr(\|\mathbf{M}\boldsymbol{\varepsilon}\| > K) \leq p\sigma^2/K^2$ for every K , and therefore $\delta_n = \max_i |d_i| \leq (\max_i h_{ii})^{1/2} \|\mathbf{M}\boldsymbol{\varepsilon}\| \rightarrow 0$ in probability. Let $A_n = n^{-1} \sum_i \exp(\varepsilon_i)$, which tends to $E e^\varepsilon$ in probability by the weak law of large numbers. Because $\hat{\varepsilon}_i = \varepsilon_i - d_i$, we have

$$e^{-\delta_n} A_n \leq \frac{1}{n} \sum_{i=1}^n \exp(\hat{\varepsilon}_i) \leq e^{\delta_n} A_n,$$

and both bounds tend to $E e^\varepsilon$ in probability. This proves [Eq. 22.5.2](#). The last statement follows because $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) \rightarrow \exp(\mathbf{x}_0^\top \boldsymbol{\beta})$ in probability, and products of convergent sequences converge. ■

By part (a), a transformed model is naturally read through medians: $h(\mathbf{x}^\top \hat{\boldsymbol{\beta}})$ estimates the conditional median when the errors are symmetric. Part (b) is the **retransformation bias**: for the exponential, $h(\mathbf{x}^\top \hat{\boldsymbol{\beta}})$ underestimates the mean by the factor $E e^\varepsilon$. Under normal errors this suggests $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + s^2/2)$, but the factor depends on the whole error distribution, and with nonnormal errors that estimate is in general inconsistent: it converges to $\exp(\sigma^2/2)$, not to $E e^\varepsilon$. Part (c) is Duan’s (1983) remedy, the empirical distribution of the residuals standing in for F , under the familiar condition that no observation keep a fixed share of the fit ([Section 6.8](#)). For general h the smearing estimate is $n^{-1} \sum_i h(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + \hat{\varepsilon}_i)$ ([Exercise \(C1\)](#)).

Example. (Skewed errors)

The script simulates the log-linear model $\log Y = 1 + 0.5x + \varepsilon$ with $n = 100$ equally spaced values of x in $[0, 2]$ and right-skewed errors $\varepsilon = 0.6(E - 1)$, E exponential with mean 1. The errors have mean 0 and variance 0.36. The true factor is $E e^\varepsilon = e^{-0.6}/0.4$, about 1.372, while the normal-theory factor $\exp(0.36/2)$ is about 1.197. At $x_0 = 1$, averaged over 4000 samples, the ratio of the estimate to the true mean is 0.730 for the naive $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})$, 0.877 for the normal-theory correction and 0.986 for the smearing estimate. Its remaining bias of about one percent reflects that residuals vary slightly less than errors.

```
import numpy as np

def simulate(reps, n_sim, seed):
    """Log-linear model with right-skewed errors  $e = 0.6 (E - 1)$ ,  $E \sim \text{Exp}(1)$ :
    estimate / true mean."""
    rng = np.random.default_rng(seed)
    xs = np.linspace(0, 2, n_sim)
    Xs = np.column_stack([np.ones(n_sim), xs])
    beta, sig = np.array([1.0, 0.5]), 0.6
    x0 = np.array([1.0, 1.0])
    true_mean = np.exp(x0 @ beta) * np.exp(-sig) / (1 - sig)  #  $E \exp(e) = e^{-(0.6-0.36)}$ 
```

```

sig)/(1 - sig)
out = np.empty((reps, 3))
for r in range(reps):
    e = sig * (rng.exponential(size=n_sim) - 1)
    z = Xs @ beta + e
    coef, *_ = np.linalg.lstsq(Xs, z, rcond=None)
    res = z - Xs @ coef
    s2_sim = res @ res / (n_sim - 2)
    naive = np.exp(x0 @ coef)
    out[r] = [naive, naive * np.exp(s2_sim / 2), naive * np.mean(np.exp(res))]
return out.mean(axis=0) / true_mean

ratios = simulate(reps=300, n_sim=100, seed=2205)
print("mean of estimate / true mean:  naive %.3f   lognormal %.3f   smearing %.3f" %
      tuple(ratios))

```

[Run this code online](#)

The browser cell uses 300 samples, so its ratios differ slightly from those quoted.

Warning. Smearing needs identically distributed errors on the transformed scale. If their variance depends on the regressors, so does $E e^\varepsilon$, and a single average corrects too much where the variance is small and too little where it is large ([Exercise \(C2\)](#)); Manning (1998) showed how large the errors can be. Check the residual spread on the transformed scale ([Chapter 21](#)) before retransforming a mean.

22.5.2 Intervals on the original scale

Proposition 22.5.2 — Intervals after a transformation

Assume [Eq. 22.5.1](#), and let Y_0 be a new response at $\mathbf{x}_0$, independent of $\mathbf{Y}$ and following the same model.

- (a) If $L \leq U$ are statistics with values in the range of g such that $\Pr\{L \leq g(Y_0) \leq U\} = 1 - \alpha$, then $\Pr\{h(L) \leq Y_0 \leq h(U)\} = 1 - \alpha$. Under normal errors the prediction interval of Theorem 12.3.1 therefore transforms into an exact prediction interval for Y_0 .
- (b) If $[L, U]$ is a confidence interval for $\mathbf{x}_0^\top \boldsymbol{\beta}$ with coverage $1 - \alpha$ and F has median 0, then $[h(L), h(U)]$ is a confidence interval for the median of Y_0 with coverage $1 - \alpha$.
- (c) Let $g = \log$, let the errors be normal, and let $\mathbf{X}$ have rank $r < n$ with $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ and $h_0 = \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^- \mathbf{x}_0$. The logarithm of the mean, $\theta = \mathbf{x}_0^\top \boldsymbol{\beta} + \sigma^2/2$, is estimated by $\hat{\theta} = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + s^2/2$, which is unbiased with

$$\text{Var}(\hat{\theta}) = \sigma^2 h_0 + \frac{\sigma^4}{2(n-r)}. \quad (22.5.3)$$

Proof.

- (a) and (b): since g is strictly increasing with inverse h , the events $\{L \leq g(Y_0) \leq U\}$ and $\{h(L) \leq Y_0 \leq h(U)\}$ are equal, and so are $\{L \leq \mathbf{x}_0^\top \boldsymbol{\beta} \leq U\}$ and $\{h(L) \leq$

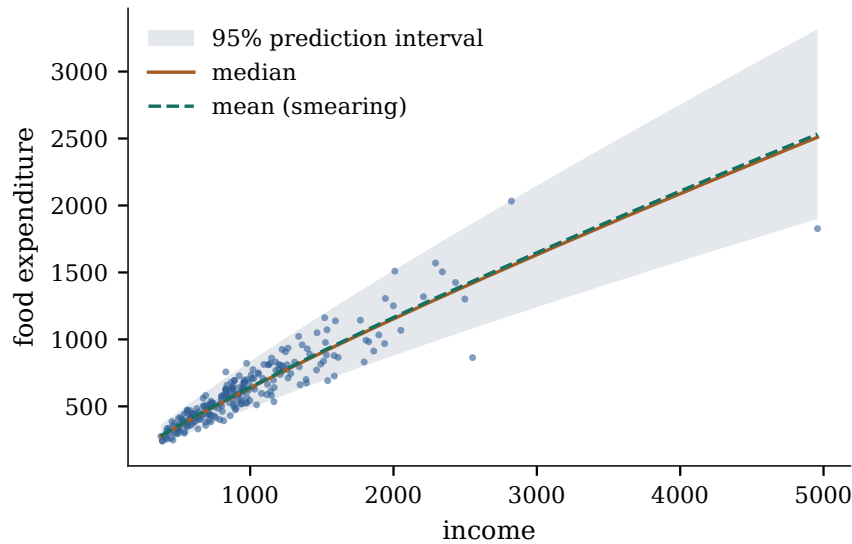


Figure 22.5.1: Engel's data on the original scale with the log-log fit transformed back: the median curve $\exp(\hat{\beta}_0 + \hat{\beta}_1 \log x)$, the smearing estimate of the mean (dashed), nearly indistinguishable here because s^2 is small, and the 95% prediction band, which widens and becomes asymmetric as income grows.

$h(\mathbf{x}_0^\top \boldsymbol{\beta}) \leq h(U)\}$; by Proposition 22.5.1(a), $h(\mathbf{x}_0^\top \boldsymbol{\beta})$ is the median of Y_0 . (c) By Theorem 7.5.1, $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ is normal with mean $\mathbf{x}_0^\top \boldsymbol{\beta}$ and variance $\sigma^2 h_0$, $(n-r)s^2/\sigma^2 \sim \chi^2(n-r)$, and the two are independent. So $E\hat{\theta} = \theta$, $\text{Var}(s^2) = 2\sigma^4/(n-r)$, and the variances add.

■

Part (c) gives the approximate interval $\exp\{\hat{\theta} \pm z_{\alpha/2}(s^2 h_0 + s^4/\{2(n-r)\})^{1/2}\}$ for the mean, approximate because the variance is estimated and $\hat{\theta}$ is only approximately normal, and dependent on the lognormal assumption. The intervals of parts (a) and (b) are exact under the normal model on the log scale.

Example. (Engel's data on the original scale)

With the log-log fit of Example (Food expenditure on the raw and the log scale), Figure 22.5.1 shows the data on the original scale with the fitted median curve, the smearing estimate of the mean and a 95% prediction band. For a household with income 500 the estimated median food expenditure is 352.2, with 95% confidence interval [342.2, 362.5], and a 95% prediction interval for a single household (Theorem 12.3.1 and Proposition 22.5.2(a)) is [269, 462]. At income 2000 the median is 1153.7, with interval [1111.7, 1197.3], and the prediction interval is [879, 1514]. A fraction 0.455 of the households lie below the fitted median curve.

The log-scale residual variance $s^2 = 0.01871$ is small, and so is the retransformation bias: the smearing factor is 1.0091 and $\exp(s^2/2)$ is 1.0094. The estimated mean at income 2000 is 1164.2 by smearing and 1164.5 under normality, about one percent above the median. The approximate 95% interval for the mean from Eq. 22.5.3 is [1122.4, 1208.3] at income 2000 and [345.4, 365.9] at income 500.

```

import numpy as np
import statsmodels.api as sm
from scipy import stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), np.log(income)])
fit = sm.OLS(np.log(food), X).fit()
b, s2, resid = fit.params, fit.scale, fit.resid
XtX_inv = np.linalg.inv(X.T @ X)
t = stats.t.ppf(0.975, n - 2)

smear = np.mean(np.exp(resid))                # Duan's smearing factor
lognormal = np.exp(s2 / 2)                    # the normal-theory factor
print(f"smearing factor {smear:.4f}, lognormal factor {lognormal:.4f}")

for x0 in (500, 2000):
    v0 = np.array([1, np.log(x0)])
    eta, h0 = v0 @ b, v0 @ XtX_inv @ v0
    median = np.exp(eta)
    ci_med = np.exp(eta + np.array([-1, 1]) * t * np.sqrt(s2 * h0))      # for
    the median, exact
    pi = np.exp(eta + np.array([-1, 1]) * t * np.sqrt(s2 * (1 + h0)))    # for a
    new household, exact
    print(f"income {x0}: median {median:.1f}, CI [{ci_med[0]:.1f}, {ci_med[1]:.1f}];"
          f" mean {median * smear:.1f} (smearing), {median * lognormal:.1f} ("
          f" lognormal);"
          f" 95% prediction interval [{pi[0]:.0f}, {pi[1]:.0f}]")

```

[Run this code online](#)

22.5.3 Elasticities and marginal effects

In a transformed model a coefficient is not a rate of change of y per unit of x ; rates and elasticities follow by differentiating through both transformations.

Proposition 22.5.3 — Elasticities in a transformed model

Let $x_1 > 0$, and suppose that the median $m(\mathbf{x})$ of Y satisfies $m(\mathbf{x})^{(\lambda)} = \beta_0 + \beta_1 x_1^{(\alpha)} + c(\mathbf{x}_{-1})$, where c does not involve x_1 and the Box–Cox transformation Eq. 22.2.1 is applied to both m and x_1 . Then

$$\frac{\partial m}{\partial x_1} = \beta_1 x_1^{\alpha-1} m^{1-\lambda}, \quad \frac{x_1}{m} \frac{\partial m}{\partial x_1} = \beta_1 x_1^\alpha m^{-\lambda}. \quad (22.5.4)$$

In particular the elasticity is the constant β_1 in the log–log model ($\lambda = \alpha = 0$), $\beta_1 x_1$ in the log–level model ($\lambda = 0$, $\alpha = 1$), β_1/m in the level–log model ($\lambda = 1$, $\alpha = 0$) and $\beta_1 x_1/m$ in the untransformed model. Under Eq. 22.1.1 the mean is a constant multiple of the median, so for $\lambda = 0$ the elasticity of the mean is the same.

Proof. Differentiate the identity with respect to x_1 . By Proposition 22.2.1(b) the derivative of $m^{(\lambda)}$ with respect to m is $m^{\lambda-1}$ and that of $x_1^{(\alpha)}$ is $x_1^{\alpha-1}$, so $m^{\lambda-1} \partial m / \partial x_1 = \beta_1 x_1^{\alpha-1}$. Multiply by $m^{1-\lambda}$, and then by x_1/m . If the mean is $k m(\mathbf{x})$ for a constant k , the factor k cancels in the elasticity. ■

The model fixes the *shape* of the elasticity profile before the data have a say. For Engel's data the log-log slope 0.8559 (interval [0.816, 0.896]) is one elasticity for all households. The untransformed linear fit implies an elasticity $\hat{\beta}_1 x / \hat{m}$ rising from 0.62 at income 500 to 0.87 at income 2000, only because a line with a positive intercept forces that shape. The Box-Tidwell estimate $\hat{\alpha} = -0.182$ of Example (Transforming income), with $\lambda = 0$, implies an elasticity proportional to $x^{-0.182}$, which *falls* between the same incomes by the factor $4^{-0.182} = \exp(-0.182 \log 4)$, about 0.78. Which shape the data support is a question about the mean function (Section 22.3).

22.5.4 Which scale to report

A transformation chosen for statistical reasons need not be the reporting scale. The logarithm is often worth keeping, since its coefficients are ratios and elasticities (Proposition 22.1.1). For other powers, report on the original scale: medians and prediction intervals, which transform exactly, and means, which need a correction factor and a constant variance. Compare models fitted on different scales on a common scale, through the Box-Cox likelihood (Section 22.2) or the accuracy of predictions of y (Chapter 29), not through R^2 (Exercise (B3)).

22.5.5 Exercises

22.5.5.1 A. Check your understanding

Exercise A1.

For Engel's log-log fit $s^2 = 0.01871$. Check that $\exp(s^2/2)$ is 1.0094. The smearing factor is 1.0091. Why are the two so close here, and in what kind of data would they differ substantially?

Exercise A2.

Explain why the prediction interval of Example (Engel's data on the original scale), obtained from Theorem 12.3.1, is longer above the median than below it, and why its coverage is nevertheless exactly 95% under the normal model on the log scale.

22.5.5.2 B. Practice

Exercise B1.

Under normal errors on the log scale, show that $E \exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) = \exp(\mathbf{x}_0^\top \boldsymbol{\beta} + \sigma^2 h_0/2)$, so the naive median estimate is biased upwards by a factor that tends to one as $h_0 \rightarrow 0$. Suggest an approximately mean-unbiased alternative, and explain why $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})$ is

exactly median-unbiased.

Solution. $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \sim N(\mathbf{x}_0^\top \boldsymbol{\beta}, \sigma^2 h_0)$ (Theorem 7.5.1), and the normal moment generating function (Theorem 3.2.1) at 1 gives the mean. The median of $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ is $\mathbf{x}_0^\top \boldsymbol{\beta}$, and exp preserves medians, so $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})$ is exactly median-unbiased for $\exp(\mathbf{x}_0^\top \boldsymbol{\beta})$. For mean-unbiasedness one can use $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} - s^2 h_0/2)$, which is approximately unbiased. The factor $\exp(\sigma^2 h_0/2)$ is tiny when h_0 is small, so the choice rarely matters.

Exercise B2.

Let $\sqrt{Y_i} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i$ with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$. Show that $E(Y | \mathbf{x}_0) = (\mathbf{x}_0^\top \boldsymbol{\beta})^2 + \sigma^2$ and that the smearing estimate $n^{-1} \sum_i (\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + \hat{\varepsilon}_i)^2$ equals $(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})^2 + \text{SSE}/n$. How does this compare with the plug-in estimate $(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})^2 + s^2$?

Exercise B3.

For Engel's data at income 500, $s^2 = 0.01871$, $h_0 = 0.01147$ and $n - r = 233$. Evaluate the two terms of Eq. 22.5.3 with σ^2 replaced by s^2 , and check the standard error 0.0147 used for the interval [345.4, 365.9]. Which term dominates, and why?

22.5.5.3 C. Going deeper

Exercise C1.

Extend Proposition 22.5.1(c) to a general strictly increasing h that is continuously differentiable with $|h'(t)| \leq K(1 + e^{\kappa|t|})$ for some constants, assuming $E e^{\kappa'|\varepsilon|} < \infty$ for some $\kappa' > \kappa$. Show that $n^{-1} \sum_i h(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + \hat{\varepsilon}_i)$ converges in probability to $E(Y | \mathbf{x}_0)$. (Use the mean value theorem and the bound on $\max_i |\hat{\varepsilon}_i - \varepsilon_i|$ from the proof.)

Exercise C2.

Suppose that on the log scale $\varepsilon_i \sim N(0, \sigma^2(\mathbf{x}_{(i)}))$, independent, with the variance depending on the regressors. Show that $E(Y | \mathbf{x}_0) = \exp\{\mathbf{x}_0^\top \boldsymbol{\beta} + \sigma^2(\mathbf{x}_0)/2\}$, and that, when the leverages tend to zero and the average $n^{-1} \sum_i \exp\{\sigma^2(\mathbf{x}_{(i)})/2\}$ converges to a limit $\bar{c}$, the smearing factor tends to $\bar{c}$. For which $\mathbf{x}_0$ does the smearing estimate overestimate the mean, and for which does it underestimate it?

Solution. Given $\mathbf{x}_0$, $\log Y$ is normal with mean $\mathbf{x}_0^\top \boldsymbol{\beta}$ and variance $\sigma^2(\mathbf{x}_0)$, and the moment generating function (Theorem 3.2.1) gives the mean. The argument of Proposition 22.5.1(c) goes through with $A_n = n^{-1} \sum_i e^{\varepsilon_i}$, whose mean is $n^{-1} \sum_i \exp\{\sigma^2(\mathbf{x}_{(i)})/2\}$; if the variances are bounded, its variance tends to zero, so $A_n \rightarrow \bar{c}$. The smearing estimate therefore targets $\exp(\mathbf{x}_0^\top \boldsymbol{\beta}) \bar{c}$. It overestimates the mean where $\exp\{\sigma^2(\mathbf{x}_0)/2\} < \bar{c}$, that is, where the log-scale variance is below its average in this sense, and underestimates it where the variance is above.